A Shrinkage Law for Epistemic Transport at the Writing Step of Language-Model Pipelines

with an open measurement kit and two parameter cards

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Draft v0.1 — August 2026

Abstract

When a language model writes the final answer of a multi-component pipeline, the epistemic qualifications raised upstream—disclosed assumptions, staleness warnings, jurisdictional distinctions, conditions of validity—do not transport into the output. Across 1,340 audited runs of two architectures sharing one writer model, output qualification follows a single line: $y = ws + c$, the form of a shrinkage estimator that weighs upstream supply s against a prior. Under a conditional-preservation instruction the evidence weight is $w = 0.35$ $[0.31, 0.39]$; under the canonical mixture-of-agents prompt it is $w = 0.16$ $[0.08, 0.23]$, statistically inseparable from ignoring upstream entirely; and compensating invention at zero supply—qualification fabricated when no component raised any—appears in both architectures (40% and 28% of zero-supply runs). Three further results sharpen the law: the parameters are set by the writing instruction, not the architecture; corroboration of a claim by multiple components does not raise its transport on our validated measurement channel; and no intervention we tested (four instructions, three training objectives, one correction loop) moved the parameters durably, while post-hoc selection with a verifier the generator never observes reliably does. We release a zero-dependency measurement kit that estimates the parameters from any pipeline's logs, carrying the identifiability guards that disqualified 21 of our own 27 experimental arms and the instrument-validation protocol without which our own second instrument measured nothing. Finally we derive the *optimal* parameters from upstream reliability, showing that perfect transport (w=1, c=0) is optimal only for perfectly reliable components: the parameters are diagnostics for choosing interventions, never objectives to train or search against.

1 Introduction

Multi-component language-model systems end, almost universally, in a writing step: one model reads what specialists, proposers, or retrieved documents produced and writes the answer a user sees. A large literature measures whether this step preserves *accuracy*. This paper measures whether it preserves *epistemic content*—the assumptions a component disclosed, the staleness it warned about, the conditions under which its claim holds. These are the parts of an answer that tell a reader how much to trust the rest, and they are precisely the parts a downstream reader cannot reconstruct if the writer drops them.

Our central finding is that the writing step does not transport this content. It behaves as an estimator: it emits the level of qualification an answer of its type ought to carry, weighing the upstream evidence against a prior learned long before the pipeline existed. That behavior is captured by one measurable law, and the law by two parameters,

$$y = w\,s + c, \tag{1}$$

where s is the number of qualification families the upstream components raised, y the number in the final output, w the *evidence weight*, and c the *prior fill*. A transducer has $w = 1$, $c = 0$. A writer that ignores its inputs has $w = 0$. Every system we measured sits low in between, and the two corollaries of sitting there are the phenomena a deployer actually experiences: at high supply the writer discards most of what its components raised, and at zero supply it *invents*—emitting qualifications with no upstream source, at rates we measure at 28% and 40% of zero-supply runs in the two architectures.

The contributions are:

1. **The law, measured** (§4): parameter cards for two architectures sharing one writer—a routed specialist council and a canonical mixture-of-agents (MoA) stack—with bootstrap intervals, a five-level supply sweep in which zero supply is measured rather than extrapolated, and per-family decomposition on a validated channel.

2. **What sets the parameters** (§5): the writing instruction moves w more than the entire architectural difference ($0.16 \to 0.35$); the compensation phenomenon travels across architectures; and corroboration—the one reliability signal a multi-agent system uniquely produces—does not modulate transport on the validated channel.

3. **Interventions under the law** (§6): instructions, preference training, and detector-fed correction loops all fail, the last by producing improvement only its own instrument can see; verifier-blind selection works, and its yield must be computed from the *measured* best-of-n curve, because redraws are positively correlated and the independence formula over-promises in both architectures.

4. **The kit** (§7): a zero-dependency package that estimates (w, c), the feature-span fraction, instrument blind spots, and the clean-sample rate from logs, with identifiability guards and a per-channel instrument-validation protocol, both of which materially changed our own conclusions.

5. **The optimal parameters, derived** (§9): w^* and c^* follow from measured upstream reliability and declared error costs; $w^*{=}1$ requires perfectly reliable components, and $c^*{>}0$ is optimal whenever components miss warranted caution. The parameters are therefore diagnostics, never training or search objectives.

Scope, stated before anything else: both architectures were built in one lab, share one writer model and one authored case battery, and all rates are relative to a lexicon whose per-family validity we measured and report. The paper's claims are calibrated to that evidence, its falsification conditions are explicit (§12), and the kit exists so that any reader can produce the third card.

2 Definitions and estimators

Property class. We measure four families of epistemic qualification: disclosure of training-data staleness (*cutoff*), labeling of numbers as assumptions (*modeled*), separate treatment of jurisdictions or regimes (*jurisd*), and statements of conditions under which a claim changes (*hedging*). The framework is not specific to this class; the kit ships a second class (source attribution) as a plug-in.

Supply and emission. For one run, s is the number of families present in the text the writer actually received (verbatim capture is a precondition; see §7), and y the number present in its output. Runs shorter than 500 characters are excluded, never scored.

The regression. (w, c) are the OLS coefficients of Eq. 1 over runs, with seeded bootstrap intervals. Because y counts at most four families, the fit is reported alongside its identifiability guards: the estimate is refused when supply spans fewer than three levels, flagged as *extrapolated* when no runs at s=0 exist (an intercept fitted outside its data routinely returns the impossible $c < 0$), and flagged as *weakly identified* when the w interval exceeds 0.35. Applied to our own ledger, these guards disqualified 21 of 27 arms with $n \geq 30$—most of our own experiments were never powered for the comparisons their point estimates invited.

Per-family transmission. Family presence is binary per run, so per family the regression is unidentifiable and the registered quantities are conditional: $T_f = P(f \in \text{output} \mid f \text{ raised})$, $I_f = P(f \in \text{output} \mid f \text{ not raised})$, and the discrimination $D_f = T_f - I_f$.

Auxiliary parameters. The feature-span fraction f (share of output characters in qualification-bearing sentences) quantifies why sequence-level preference training dilutes; the clean-sample rate p_0 (probability a run contains no invented family) sizes selection (§6).

3 Instruments, validated per channel

Every number in this paper is instrument-relative, so the instruments were graded before the numbers were trusted. Against blinded dual-judge labels over 200 sentences (judge–judge agreement 0.86; accuracy 0.90/0.93 on hand-written anchors that include paraphrase positives and keyword-bait negatives), the lexicon detects the *modeled* family well (sensitivity 0.92, precision 0.92) and the remaining families conservatively (specificity $\geq$ 0.95, sensitivity 0.25–0.30). All composite rates are therefore undercounts outside the modeled family, and *modeled* is the validated channel on which per-family claims rest.

The cautionary result: our second instrument, a natural-language-inference detector with a genuine 0.93 AUC on the discrimination task it was calibrated for, was re-calibrated against the same judge labels for the task it had actually been used on—family presence—and carried *no signal*: per-family AUC 0.12–0.55, two families inversely related to truth. No threshold rescues an AUC below chance. Two instruments had "agreed" for weeks while measuring different constructs. The protocol that found this ships with the kit, and the framework's rule follows from it: a verdict requires two instruments *whose sensitivity to the construct has itself been demonstrated*; agreement between unvalidated instruments is rhetoric.

4 Two parameter cards

Table 1 is the paper's central evidence. Three readings:

The form travels. Both architectures show monotone strata, positive prior fill, and an evidence weight whose interval excludes zero: writers respond to supply, incompletely, and fill from a prior when supply is absent. Compensating invention at zero supply—the deployer-facing symptom—appears in both (40% and 28% of zero-supply runs), and on trigger-light

	Council (routed specialists, conditional instruction)	**Mixture-of-Agents** (general proposers, canonical prompt)
runs	1,260	80
evidence weight w	$0.352\ [0.310, 0.391]$	$0.158\ [0.076, 0.234]$
prior fill c	$0.540\ [0.437, 0.651]$	$0.275\ [0.104, 0.456]$
prior-trust ratio $(1-w)/w$	1.84	5.35
supply strata means (s=0..4)	.40 / .81 / 1.32 / 1.62 / 1.83	.28 / .44 / .50 / .88 / .83
strata R^2 / run-level R^2	0.976 / 0.140	0.890 / 0.136
zero-supply runs inventing	40% [29%, 52%]	27.8% [12%, 51%]
feature-span fraction f	0.068	0.002
D_{modeled} (validated channel)	$0.213\ [0.155, 0.271]$	$-0.010\ [-0.139, 0.117]$

Table 1: Parameter cards for two architectures sharing one writer model (gpt-oss-20B). Both R^2 values are reported deliberately: the law describes the conditional mean tightly and individual runs loosely, and quoting only the strata fit would oversell it. The council card pools all usable ledger runs; the MoA card is an 80-run supply sweep in which supply was manipulated by sentence-level ablation and injection of the upstream texts, so s=0 is measured, not extrapolated.

cases authored to warrant *no* qualification, so for those cases invention is miscalibration by construction, independent of any judgment about upstream quality.

The parameters do not travel. The MoA card sits near the pure-register corner: its w interval does not exclude 0.15, its prior-trust ratio is three times the council's, and on the validated channel its discrimination is indistinguishable from zero—the aggregator emits modeled qualification at the same rate whether or not any proposer raised it. Its composite slope is carried by families whose distinctive phrases the writer sometimes echoes, not by behavior on the channel we can measure cleanly. We registered the prediction that the MoA card would show the full shrinkage form with both corners excluded; it did not, and per the registration this boundary is reported at equal prominence with the successes.

The near-silent regime exists and scores deceptively well. The MoA outputs carry almost no qualification content at all ($f = 0.002$; the council's naive flat-merge arm measured $f = 0.000$). A silent writer invents little because it says little—composite invention metrics reward it, and any evaluation that does not report f beside the invention rate will mistake silence for calibration.

5 What sets the parameters

The instruction, more than the architecture. Holding the writer model fixed and restricting to a shared nine-case battery: the canonical MoA prompt over general proposers yields $w = 0.16\ [0.08, 0.23]$; a naive merge prompt over domain specialists yields $w = 0.255\ [0.13, 0.39]$; the conditional-preservation instruction over the same specialists yields $w = 0.346\ [0.22, 0.47]$. The instruction change moves w by more than the upstream-composition change, and no instruction we found moves it past roughly a third of the way to transport. Clause ablation in the earlier program localizes the effect further: a single conditional clause carries essentially the entire instruction effect, and each clause lifts primarily the family it names. Instructions act as a gain control on transport— graded, targeted, and bounded well short of fidelity.

Corroboration is not used. A multi-agent system produces one reliability signal a single model cannot: independent agreement among components. If the writer performed any reliability weighting, a family raised by two or more components should transport at a higher rate than a family raised by one. On the validated channel it does not: $T(k{=}1) = 0.524$ ($n{=}824$) versus $T(k{\geq}2) = 0.487$ ($n{=}76$), difference CI $[-0.153, +0.080]$. The three low-validity families show large monotone agreement effects (up to +0.43), which we report without a mechanism claim: genuine family-specific weighting, phrase-exposure artifact (more components using a distinctive phrase mechanically raises the chance one detectable phrasing survives), and case-mix confounding are all live, and separating them requires judge labels rather than a lexicon. On the channel we can trust, the summary of this section is one sentence: *instructions set the gain; nothing we found sets the calibration*— not architecture, not training (§6), and not the architecture's own agreement structure.

6 Interventions under the law

The law explains why the obvious interventions fail, and the failures in turn support the law. We summarize; the companion behavioral paper reports each in full.

Instructions move surface density (the gain of §5) but leave invention intact: an explicit suppression clause left zero-supply invention statistically unchanged. Under the law this is expected—shrinkage toward a prior is loss-optimal on the training distribution, and no instruction changes what is loss-optimal.

Sequence-level preference training (DPO/ORPO variants, three objective classes) produced the dissociation the feature-span fraction predicts: preference accuracy rose $0.48 \to 0.94$ while emitted behavior did not move. With $f \approx 0.07$, roughly 93% of a sequence-level margin is satisfiable off-feature; we measured the pair construction used in the failed runs at 0.87 off-feature mass, against 0.00 for programmatically constructed minimal pairs.

Detector-fed correction is worse than nothing. A runtime audit that quoted its detector's matched phrases back to the writer removed 88% of flagged inventions *by that detector's count*; manual inspection showed the qualifications surviving in rewording, and one flagged "invention" was the detector's own false positive. The instrument that generates feedback can never grade compliance—and a second instrument only helps if its sensitivity is demonstrated (§3).

Verifier-blind selection is the intervention the law permits. Sample n outputs, keep the one whose provenance audit is cleanest, never reveal the verifier to the generator; annotate residual failures rather than revising them. Structurally, selection conditions the output distribution while feedback conditions the policy input; only the latter can be gamed, because only the latter is observable. Sizing must use the *measured* curve: in both architectures the empirical best-of-n yield sits below the independence formula $1-(1-p_0)^n$ (council: 0.83 measured vs. 0.94 predicted at $n{=}3$; MoA: 0.78 vs. 0.89 at $n{=}2$), because a writer that invents on a prompt tends to invent again on redraws (corpus-level ICC 0.190; §8). The kit computes both curves and reports how many distinct tasks support each point.

7 The measurement kit

The kit (`gst-kit`, MIT, zero dependencies in the core) turns a pipeline's logs into a parameter card. Integration is one adapter from the system's log shape to a four-field record—upstream texts as the writer received them, output, task identifier, condition—with prebuilt adapters for MoA stacks, LangGraph- and AutoGen-style frameworks, retrieval pipelines, and generic

JSONL. Three design commitments came from measurement failures documented in our own program:

1. **Guards over estimates.** Every estimator refuses or flags an answer its input could not identify (supply that never varies, an extrapolated intercept, a weakly identified slope, best-of-n points supported by an unrepresentative subpopulation). Each guard exists because its absence produced a wrong number that we published internally or nearly published.

2. **Execution-path assertion.** The writer prompt actually used is captured and asserted per run; violating runs are quarantined and counted. Thirty-nine of our runs silently measured a prompt that never executed, and two verdicts had to be withdrawn.

3. **Structural safety over procedural care.** The selection API takes a zero-argument sampler; passing a callable that could receive feedback is a type error, not a guideline.

8 Measured regularities across the corpus

Beyond the two cards, the full corpus (1,300 usable runs, 25 experimental arms meeting the inclusion rule) supports corpus-level estimates of three regularities, each pre-registered with its bar checked for attainability at the realized n. A fourth registered prediction falsified and is reported here at the same prominence.

The evidence weight is bounded away from transduction, with real between-arm spread. A random-effects meta-analysis over all qualifying arms—weakly identified arms included with their wide standard errors, which is the point of the machinery—gives pooled $\mu_w = 0.364$ $[0.290, 0.438]$ with between-arm spread $\tau = 0.158$ and a 95% prediction interval for a new arm of $[0.045, 0.682]$. No measured arm approaches faithful transduction (the highest, a trained-writer arm, reaches 0.76 with an extrapolated intercept), but the honest statement is a central tendency near 0.36 with genuine spread—not a narrow invariant band. We report this correction explicitly because an earlier informal reading of the same data claimed a tighter band than the estimate supports.

Invention is prompt-heterogeneous, quantified. Across 251 repeated (task, condition, writer) cells (1,279 runs), the intraclass correlation of invention counts is 0.190 $[0.128, 0.242]$: roughly a fifth of invention variance lies between prompts. Within-cell counts are mildly under-dispersed (median variance-to-mean 0.93), consistent with near-binary counts of bounded families rather than a Poisson process. This is the corpus-level basis for the selection-sizing correction of §6: redraws are correlated because rates differ by prompt, so the independence formula over-promises.

The prior fill is a property of the output register. Pooling fitted intercepts by the register the writer emits into: prose-register arms give $c = 0.407$ $[0.164, 0.651]$ (heterogeneous, $\tau = 0.270$); the two arms writing under an attribution-table protocol give $c = -0.023$ $[-0.110, 0.065]$ with $\tau = 0.000$—the two arms agree exactly—and the intervals are disjoint. A decision-formatted arm sits highest at 0.805 (single arm, descriptive). Invention, in other words, is not set by the task but by the textual register the answer is written in; registers whose conventions demand per-item attribution show no measurable prior fill. The mechanism-level and intervention-level claims about attribution-constrained writing remain under registration and are deliberately not made here.

The falsified sibling. We registered that the validated modeled channel carries less than its equal share of the composite slope (aggregation distortion); the measured share is 0.219

$[0.170, 0.268]$, whose interval includes the equal share 0.25. The claim is withdrawn from this paper's argument. Descriptively, the FP-prone hedging channel carries the largest single share ($\sim 39\%$), so composite slopes should still be read beside per-family validity— but as caution, not as a demonstrated distortion.

9 The optimal parameters are derived, not chosen

It is tempting to read Table 1 as a scoreboard on which $w{=}1$, $c{=}0$ wins. It is not, and the reason disciplines the whole framework. The writer faces a signal-detection problem: upstream supply is a noisy signal of whether qualification is *warranted*. The optimal response to a noisy signal is exactly a shrinkage estimator, with

$$w^* \;=\; \frac{\tau^2}{\tau^2 + \sigma^2}, \tag{2}$$

the upstream components' signal-to-noise ratio. Perfect transport is optimal only for perfectly reliable components. Our specialists are not: they raise qualification on content authored to warrant none, so $w^* < 1$; and to the extent they *miss* warranted caution, $c^* > 0$—some invention is a writer correctly filling gaps. The measured pathology is therefore not shrinkage itself but *miscalibrated* shrinkage: weights set by pretraining priors rather than by the measured reliability of the actual upstream, and invention uncorrelated with warrant (the zero-supply inventions occur on cases authored to warrant nothing).

Two consequences. First, the honest scope of our normative claims: "the writer underweights its components" presumes the components deserve the weight, and upstream warrant-precision is a quantity we have not measured; our unconditional claims are anchored on the authored zero-warrant cases. Second, and centrally: because (w^*, c^*) are functions of measured upstream reliability and *declared* error costs, legitimate optimization targets the derivation—never the diagnostic. Training or searching against (w, c) directly selects for transcription and for instrument artifacts (a perfect score describes a parrot, not a judge), and at ecosystem scale a reported diagnostic becomes a target without anyone optimizing it deliberately. The framework's rule: measure (w, c) to choose an intervention; estimate (w^*, c^*) from held-out reliability measurements and stated costs if a target is needed; and keep evaluation batteries sealed from tuning.

10 Related work

Mixture-of-agents aggregation was introduced by Wang et al. [1]; Li et al. [2] showed aggregation quality is sensitive to proposer quality, the accuracy-domain analogue of our reliability argument. A 2026 literature documents uncertainty loss in agent pipelines: Shi et al. [3] name the phenomenon of fragile upstream states repackaged as confident artifacts, and propagation frameworks [4, 5, 6] monitor confidence across components. Relative to this line, our contributions are (i) *bidirectionality*—we measure invention under scarcity, not only loss; (ii) a *quantitative law* with estimable parameters rather than a qualitative failure mode; (iii) an *operational calibration test* for aggregators (fitted w against derived w^*); and (iv) the instrument-validation protocol without which, our own experience shows, such measurements are unreliable. Single-model neighbors measure certainty distortion between input and output [7] and uncertainty preservation in clinical summarization [8]; neither addresses the multi-component writing step or supplies portable measurement. The learning–acting dissociation under sequence-level preference objectives connects to known credit-assignment limits of DPO-family losses [9, 10]; Goodhart dynamics under optimization pressure are classical [11, 12] and we treat our detector-fed correction result as a measured instance.

11 Limitations

Both cards come from one lab, one writer model, one authored nine-case battery, and one property class; the MoA card is corroboration, not independent replication. All rates are lexicon-relative; outside the modeled family the lexicon is a high-precision undercounter, and per-family claims are made only on the validated channel. Supply in the MoA sweep was manipulated by ablation and injection, so its high-supply strata measure mixed upstream. Upstream warrant-precision is unmeasured, which bounds the normative reading of the discard results (§9); the invention results are anchored on authored zero-warrant cases and do not share that bound. w and D_f are different estimands; we compare them only where registered. The agreement-effect mechanism on low-validity families is unresolved. All experiments were pre-registered with consequences stated in advance; registrations, amendments, and two corrections issued during this program's own audit are in the public runbook.

12 Falsification and replication

The framework is refuted, not embarrassed, by: a writer trained on ordinary text showing $w \approx 1$, $c \approx 0$ without intervention; a supply response no monotone estimator reading survives; sequence-level preference training moving a small-f behavior at modest scale; or selection showing improvement under its own verifier that an independent validated instrument denies. The cleanest single test is the selection-versus-feedback asymmetry: the same verifier, used post-hoc versus in-loop, is predicted to produce real versus instrument-relative improvement. Running any of these requires an afternoon: one adapter, one `gst measure` invocation, and a supply sweep by ablation if natural variation is insufficient. The third parameter card—anyone's—is the most useful thing this paper can cause to exist.

13 Conclusion

One line describes what a pipeline's writing step does to epistemic content; two numbers describe how badly; both are measurable from logs in an afternoon. In every configuration we built, the writer weighed its own prior above its components' evidence, filled silence with invented caution on cases authored to warrant none, ignored the one reliability signal its architecture uniquely produced, and resisted every correction except being sampled and selected by a verifier it never saw. The instruction is the only dial that moves the evidence weight, and it stops a third of the way to fidelity. Whether these parameters are constants of current models or artifacts of our corner of the design space is exactly the question the kit exists to settle.

References

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